

Hector Manuel Lugaro Rodriguez

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EDUCATION

Massachusetts Institute of Technology

Bachelor of Science in Engineering

Cambridge, MA

Class of 2026

- **GPA:** 4.9/5.0
- **Relevant Coursework:** Graduate Robotic Manipulation, Engineering Systems Design & Development II, Electronics for Mechanical Systems II, Design & Manufacturing II, Dynamics & Control II

RESEARCH EXPERIENCE

Japan Agency for Marine-Earth Science and Technology (JAMSTEC)

Deep-Sea Hydrodynamics Researcher – FF11K Marine Lander

Yokohama, Japan

Summer 2025

- Simulated extreme deep-sea hydrodynamics using OpenFOAM to predict descent trajectories and terminal velocities for an 11km-rated modular marine lander.
- Analyzed fluid dynamics and drag coefficients to optimize the mechanical geometry, ensuring hydrodynamic stability during unpowered deep-ocean deployment.
- Modeled high-pressure aquatic environments to validate the lander's structural and hydrodynamic performance prior to physical prototyping.

Engineering Faculty, Pan-American University

Research Assistant – Chameleon-Inspired Robotics

Mexico City, CDMX

Summer 2024

- Engineered Python-based YOLO object recognition and a PID control algorithm for autonomous target tracking using a Raspberry Pi 5 and ArduCAM PTZ module.

ENGINEERING PROJECTS

Remora ROV: Autonomous Inspection Vehicle | *Hardware Integration & Autonomy*

- Led the electromechanical retrofitting, subsea power logistics, and neutral buoyancy trimming for an 8-thruster, 300-meter rated underwater vehicle.
- Engineered topside-to-subsea communications using a Fathom-X interface to transmit MAVLink commands over a custom high-voltage tether.
- Developed software-in-the-loop (SITL) autonomy architecture using Gazebo and ArduSub to simulate and execute complex orbital flight maneuvers prior to deployment.

Adversarial Ping Pong System | *Robotic Manipulation*

- Engineered an adversarial robotic table tennis system utilizing a Kuka LBR iiwa robotic arm, focusing on dynamic target interception and rapid kinematic response.
- Implemented real-time trajectory prediction, inverse kinematics, and IRIS regions for collision-free path planning using the Drake simulation framework.
- Collaborated on the formal system architecture and algorithm validation using Deepnote to optimize the arm's striking accuracy and joint-space limits.

Autonomous Plastic Sorting Robotics | *Electromechanical Systems*

- Designed and integrated a robotic system aimed at identifying and sorting plastic pollution, bridging physical mechanical handling with automated classification logic.

SKILLS & INTERESTS

Software & Tools: Python, C, MATLAB, Drake, OpenFOAM, KiCad, ROS, L^AT_EX, Git

Hardware & Manufacturing: CNC Machining, Soldering, SLA & FFF, Injection Molding, PCB Layout

Languages: English (Native), Spanish (Native), Japanese (JLPT N3 Certified)

Interests: Underwater Robotics, Marine Systems Integration, Machine Learning, Marathon Training